

Assignment 3: File Handling

Operating System Lab (Unix)

Problem Statement

Write C programs to perform file operations using low-level system calls in a Unix/Linux environment. The objective is to understand how files are created, accessed, modified, and managed at the system level.

Tasks to Implement

1. Write a program to create a file using `creat()` and display a success message
2. Write a program to open a file using `open()` and display file descriptor
3. Write a program to write data into a file using `write()`
4. Write a program to read contents of a file using `read()` and display on screen
5. Write a program to copy contents from one file to another
6. Write a program to demonstrate file pointer movement using `lseek()`
7. Write a program to duplicate file descriptor using `dup()`

Instructions

- Use C programming language with system calls only
- Compile using `gcc filename.c -o output`
- Execute using `./output`
- Use headers: `unistd.h`, `fcntl.h`, `sys/types.h`, `sys/stat.h`
- Handle errors using return values (e.g., `-1`)
- Display meaningful messages for each operation

Sample Output

Example (write):

```
File created successfully.  
Data written to file.
```

Example (read):

```
File contents:  
Hello, Operating Systems!
```

Constraints

- Use only system calls (no `fopen()`, `fprintf()`, etc.)
- Programs must compile and run without errors
- Ensure proper closing of files using `close()`
- Avoid hardcoding file paths (use user input where possible)

Evaluation Criteria

- Correctness of File Operations – 40%
- Understanding of System Calls – 20%
- Error Handling – 15%
- Code Clarity and Structure – 15%
- Documentation / Comments – 10%

Submission Guidelines

- Submit all programs in a single compressed (.zip) file
- Folder name: `RollNo_Assignment3`
- Include source files and sample input/output
- Include a README explaining each program

Bonus Tasks (Optional but Recommended)

1. Write a program to count number of characters, words, and lines in a file
2. Implement file copy with buffer (efficient version)
3. Demonstrate use of `dup2()` for redirecting output to a file
4. Write a program to append data to an existing file
5. Create a simple logging system using file operations